Lab Sheet 5: SQL Joins

Use suitable tables such as Student, Department, Course, and Employee to perform the following:

1. Demonstrate an **INNER JOIN** between Student and Department.

```
156 • SELECT * FROM Student INNER JOIN Department ON Student.Department_ID = Department.Department_ID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo	Department_ID	Department_ID	Department_Name	Location
101	Anshika	Female	20	Delhi	B.Tech CSE	85.50	9876543210	101	101	IT	Delhi
102	Radha	Female	10	Kashipur	B.Tech IT	85.50	9876543276	101	101	IT	Delhi
103	Jayshree	Female	21	Delhi	B.Tech IT	85.50	9876543276	101	101	IT	Delhi
104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276	101	101	IT	Delhi
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276	101	101	IT	Delhi
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276	102	102	HR	Jaipur
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211	102	102	HR	Jaipur
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212	102	102	HR	Jaipur
109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213	102	102	HR	Jaipur
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214	102	102	HR	Jaipur
111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215	103	103	Finance	Mumbai
112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216	103	103	Finance	Mumbai
114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250	103	103	Finance	Mumbai

2. Display student names along with their department names using INNER JOIN.

```
157 • SELECT Student.Name, Department.Department_Name FROM Student INNER JOIN Department ON Student.Department_ID = Department.Department_ID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name	Department_Name
Anshika	IT
Radha	IT
Jayshree	IT
Ruhani	IT
Shreya	IT
Maya	HR
Rahul	HR
Priya	HR
Aman	HR
Neha	HR
Rohit	Finance
Simran	Finance
Aditi	Finance

3. Display employee names along with their department names.

```
158 • SELECT Employee.Name, Department.Department_Name FROM Employee INNER JOIN Department
159   ON Employee.Department = Department.Department_Name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name	Department_Name
Priya	IT
Amit	IT
Rohit	HR
Neha	Finance
Rahul	Marketing

4. Display students along with their course details.

```
160 • SELECT Student.Name,Student.Course,Course.Course_ID,Course.Fees,Course.Duration
161   FROM Student INNER JOIN Course ON Student.Course = Course.Course_Name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name	Course	Course_ID	Fees	Duration
Anshika	B.Tech CSE	1	90000.00	4 Years
Radha	B.Tech IT	3	80000.00	4 Years
Jayshree	B.Tech IT	3	80000.00	4 Years
Ruhani	B.Tech CSE	1	90000.00	4 Years
Shreya	B.Tech CSE	1	90000.00	4 Years
Maya	B.Tech CSE	1	90000.00	4 Years
Rahul	B.Tech CSE	1	90000.00	4 Years
Priya	BCA	2	60000.00	3 Years
Aman	B.Tech IT	3	80000.00	4 Years
Neha	B.Tech CSE	1	90000.00	4 Years
Rohit	BCA	2	60000.00	3 Years
Simran	B.Tech IT	3	80000.00	4 Years
Aditi	B.Tech CSE	1	90000.00	4 Years
Arjun	BCA	2	60000.00	3 Years

5. Find students belonging to a particular department using INNER JOIN.

```
162 • SELECT Student.Name, Department.Department_Name FROM Student
163   INNER JOIN Department ON Student.Department_ID = Department.Department_ID
164   WHERE Department.Department_Name = 'IT';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name	Department_Name
Anshika	IT
Radha	IT
Jayshree	IT
Ruhani	IT
Shreya	IT

6. Demonstrate a **LEFT OUTER JOIN** between Student and Departmen

```
SELECT * FROM Student LEFT JOIN Department
ON Student.Department_ID = Department.Department_ID;
```

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo	Department_ID	Department_ID	Department_Name	Location
101	Anshika	Female	20	Delhi	B.Tech CSE	85.50	9876543210	101	101	IT	Delhi
102	Radha	Female	10	Kashipur	B.Tech IT	85.50	9876543276	101	101	IT	Delhi
103	Jayshree	Female	21	Delhi	B.Tech IT	85.50	9876543276	101	101	IT	Delhi
104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276	101	101	IT	Delhi
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276	101	101	IT	Delhi
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276	102	102	HR	Jaipur
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211	102	102	HR	Jaipur
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212	102	102	HR	Jaipur
109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213	102	102	HR	Jaipur
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214	102	102	HR	Jaipur
111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215	103	103	Finance	Mumbai
112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216	103	103	Finance	Mumbai
114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250	103	103	Finance	Mumbai
141	Arjun	NULL	21	NULL	BCA	NULL	NULL	NULL	NULL	NULL	NULL

7. Display all students, including those who do not have a matching department.

```
SELECT Student.Name,Department.Department_Name
FROM Student LEFT JOIN Department ON Student.Department_ID = Department.Department_ID;
```

Name	Department_Name
Anshika	IT
Radha	IT
Jayshree	IT
Ruhani	IT
Shreya	IT
Maya	HR
Rahul	HR
Priya	HR
Aman	HR
Neha	HR
Rohit	Finance
Simran	Finance
Aditi	Finance
Arjun	NULL

8. Demonstrate a **RIGHT OUTER JOIN** between Student and Department.

```
SELECT * FROM Student RIGHT JOIN Department ON Student.Department_ID = Department.Department_ID;
```

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo	Department_ID	Department_ID	Department_Name	Location
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276	101	101	IT	Delhi
104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276	101	101	IT	Delhi
103	Jayshree	Female	21	Delhi	B.Tech IT	85.50	9876543276	101	101	IT	Delhi
102	Radha	Female	10	Kashipur	B.Tech IT	85.50	9876543276	101	101	IT	Delhi
101	Anshika	Female	20	Delhi	B.Tech CSE	85.50	9876543210	101	101	IT	Delhi
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214	102	102	HR	Jaipur
109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213	102	102	HR	Jaipur
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212	102	102	HR	Jaipur
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211	102	102	HR	Jaipur
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276	102	102	HR	Jaipur
114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250	103	103	Finance	Mumbai
112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216	103	103	Finance	Mumbai
111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215	103	103	Finance	Mumbai
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	104	Marketing	Pune
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	105	Sales	Kota

9. Display all departments, including departments having no students.

```
SELECT Department.Department_Name, Student.Name FROM Student
RIGHT JOIN Department ON Student.Department_ID = Department.Department_ID;
```

Department_Name	Name
IT	Shreya
IT	Ruhani
IT	Jayshree
IT	Radha
IT	Anshika
HR	Neha
HR	Aman
HR	Priya
HR	Rahul
HR	Maya
Finance	Aditi
Finance	Simran
Finance	Rohit
Marketing	NULL
Sales	NULL

10. Find students who are not assigned to any department using LEFT JOIN.

```
SELECT Student.Student_ID, Student.Name FROM Student LEFT JOIN Department ON
Student.Department_ID = Department.Department_ID
WHERE Department.Department_ID IS NULL;
```

Student_ID	Name
141	Arjun

11. Find departments having no students using RIGHT/LEFT JOIN.

12. Demonstrate a **FULL OUTER JOIN** concept using suitable queries/workaround where supported.

```
SELECT Student.Student_ID, Student.Name, Department.Department_Name
FROM Student LEFT JOIN Department ON Student.Department_ID = Department.Department_ID
UNION SELECT Student.Student_ID,Student.Name, Department.Department_Name
FROM Student RIGHT JOIN Department ON Student.Department_ID = Department.Department_ID;
```

Student_ID	Name	Department_Name
101	Anshika	IT
102	Radha	IT
103	Jayshree	IT
104	Ruhani	IT
105	Shreya	IT
106	Maya	HR
107	Rahul	HR
108	Priya	HR
109	Aman	HR
110	Neha	HR
111	Rohit	Finance
112	Simran	Finance
114	Aditi	Finance
141	Arjun	NULL
NULL	NULL	Marketing

13. Demonstrate a **CROSS JOIN** between Student and Course.

```
SELECT Student.Name,Course.Course_Name FROM Student CROSS JOIN Course
```

Name	Course_Name
Anshika	MBA
Anshika	MCA
Anshika	B.Tech IT
Anshika	BCA
Anshika	B.Tech CSE
Radha	MBA
Radha	MCA
Radha	B.Tech IT
Radha	BCA
Radha	B.Tech CSE
Jayshree	MBA
Jayshree	MCA
Jayshree	B.Tech IT
Jayshree	BCA
Jayshree	B.Tech CSE

14. Calculate the number of possible Student-Course combinations using CROSS JOIN.

```
SELECT COUNT(*) AS Total_Combinations FROM Student CROSS JOIN Course;
```

Total_Combinations
70

15. Demonstrate a **NATURAL JOIN** between two suitable tables.

```
SELECT * FROM Student NATURAL JOIN Department;
```

Department_ID	Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo	Department_Name	Location
101	101	Anshika	Female	20	Delhi	B.Tech CSE	85.50	9876543210	IT	Delhi
101	102	Radha	Female	10	Kashipur	B.Tech IT	85.50	9876543276	IT	Delhi
101	103	Jayshree	Female	21	Delhi	B.Tech IT	85.50	9876543276	IT	Delhi
101	104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276	IT	Delhi
101	105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276	IT	Delhi
102	106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276	HR	Jaipur
102	107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211	HR	Jaipur
102	108	Priya	Female	20	Mumbai	BCA	91.00	9876543212	HR	Jaipur
102	109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213	HR	Jaipur
102	110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214	HR	Jaipur
103	111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215	Finance	Mumbai
103	112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216	Finance	Mumbai
103	114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250	Finance	Mumbai

16. Compare the result of NATURAL JOIN and INNER JOIN.

```
SELECT * FROM Student NATURAL JOIN Department;
SELECT * FROM Student
INNER JOIN Department ON Student.Department_ID = Department.Department_ID;
```

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo	Department_ID	Department_ID	Department_Name	Location
101	Anshika	Female	20	Delhi	B.Tech CSE	85.50	9876543210	101	101	IT	Delhi
102	Radha	Female	10	Kashipur	B.Tech IT	85.50	9876543276	101	101	IT	Delhi
103	Jayshree	Female	21	Delhi	B.Tech IT	85.50	9876543276	101	101	IT	Delhi
104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276	101	101	IT	Delhi
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276	101	101	IT	Delhi
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276	102	102	HR	Jaipur
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211	102	102	HR	Jaipur
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212	102	102	HR	Jaipur
109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213	102	102	HR	Jaipur
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214	102	102	HR	Jaipur
111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215	103	103	Finance	Mumbai
112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216	103	103	Finance	Mumbai
114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250	103	103	Finance	Mumbai

17. Perform a **SELF JOIN** on an Employee table to show employee-manager relationships.

```
SELECT E.Name AS Employee, M.Name AS Manager
FROM Employee E LEFT JOIN Employee M ON E.Manager_ID = M.Employee_ID;
```

Employee	Manager
Amit	NULL
Rahul	Amit
Neha	Amit
Priya	Rahul
Rohit	Rahul

18. Display employees along with their managers using SELF JOIN.

19. Join three tables—Student, Department and Course.

```
SELECT Student.Name AS Student_Name, Course.Course_Name, Department.Department_Name
FROM Student INNER JOIN Department ON Student.Department_ID = Department.Department_ID
INNER JOIN Course ON Student.Course = Course.Course_Name;
```

Student_Name	Course_Name	Department_Name
Anshika	B.Tech CSE	IT
Radha	B.Tech IT	IT
Jayshree	B.Tech IT	IT
Ruhani	B.Tech CSE	IT
Shreya	B.Tech CSE	IT
Maya	B.Tech CSE	HR
Rahul	B.Tech CSE	HR
Priya	BCA	HR
Aman	B.Tech IT	HR
Neha	B.Tech CSE	HR
Rohit	BCA	Finance
Simran	B.Tech IT	Finance
Aditi	B.Tech CSE	Finance

20. Display student name, course name and department name using multiple joins.

```
196 •  SELECT Student.Name AS Student_Name, Course.Course_Name, Department.Department_Name
197    FROM Student INNER JOIN Department ON Student.Department_ID = Department.Department_ID
198    INNER JOIN Course ON Student.Course = Course.Course_Name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Student_Name	Course_Name	Department_Name
Anshika	B.Tech CSE	IT
Radha	B.Tech IT	IT
Jayshree	B.Tech IT	IT
Ruhani	B.Tech CSE	IT
Shreya	B.Tech CSE	IT
Maya	B.Tech CSE	HR
Rahul	B.Tech CSE	HR
Priya	BCA	HR
Aman	B.Tech IT	HR
Neha	B.Tech CSE	HR
Rohit	BCA	Finance
Simran	B.Tech IT	Finance
Aditi	B.Tech CSE	Finance

21. Find employees whose salary is greater than the average salary of their department using joins/subqueries.

```
199 •  SELECT E.Name, E.Department, E.Salary FROM Employee E
200    WHERE E.Salary > (SELECT AVG(E2.Salary) FROM Employee E2 WHERE E2.Department = E.Department);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name	Department	Salary
Priya	IT	45980.00

22. Find the department having the maximum number of students using joins and aggregation.

```
201 •  SELECT Department.Department_Name, COUNT(Student.Student_ID) AS Total_Students
202    FROM Department LEFT JOIN Student ON Department.Department_ID = Student.Department_ID
203    GROUP BY Department.Department_ID, Department.Department_Name ORDER BY Total_Students DESC LIMIT 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Department_Name	Total_Students
IT	5

23. Display departments and the total number of students in each department.

```
204 •  SELECT Department.Department_Name, COUNT(Student.Student_ID) AS Total_Students
205    FROM Department LEFT JOIN Student ON Department.Department_ID = Student.Department_ID
206    GROUP BY Department.Department_ID, Department.Department_Name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Department_Name	Total_Students
IT	5
HR	5
Finance	3
Marketing	0
Sales	0

24. Display courses along with the number of students enrolled in each course.

```
207 •  SELECT Course.Course_Name, COUNT(Student.Student_ID) AS Total_Students
208    FROM Course LEFT JOIN Student ON Student.Course = Course.Course_Name
209    GROUP BY Course.Course_ID, Course.Course_Name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Course_Name	Total_Students

25. Create a complete query using **INNER JOIN, LEFT JOIN, GROUP BY and aggregate functions** to generate a student-department-course report.

```
210 •  SELECT Department.Department_Name, Course.Course_Name,
211    COUNT(Student.Student_ID) AS Total_Students,
212    AVG(Student.Percentage) AS Average_Percentage
213    FROM Department LEFT JOIN Student ON Department.Department_ID = Student.Department_ID LEFT JOIN Course
214    ON Student.Course = Course.Course_Name GROUP BY
215    Department.Department_ID,Department.Department_Name, Course.Course_ID,Course.Course_Name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Department_Name	Course_Name	Total_Students	Average_Percentage
IT	B.Tech CSE	3	85.500000
IT	B.Tech IT	2	85.500000
HR	B.Tech CSE	3	84.083333
HR	B.Tech IT	1	74.750000
HR	BCA	1	91.000000
Finance	B.Tech CSE	1	90.500000
Finance	B.Tech IT	1	82.000000
Finance	BCA	1	69.250000
Marketing	NULL	0	NULL
Sales	NULL	0	NULL